

WEEKLY REPORT

WEEK 2

During this phase of the project, the work focused on preparing and presenting the Zeroth Review for the AI-based cardiovascular disease detection system. A presentation was compiled covering the base papers, abstract, introduction, problem statement, literature review, requirements, challenges, future scope, and conclusion of the proposed work.

Two base papers were studied and summarized, one on phonocardiogram (PCG) heart-sound classification using 1D convolutional neural networks and the other on deep transfer learning and residual network architectures for cardiac abnormality detection, to establish the foundation and relevance of the proposed system. A literature review covering related deep-learning approaches for acoustic heart-sound analysis was also prepared, identifying existing research gaps such as high dependency on subjective manual auscultation, acoustic noise interference, and the need for automated feature extraction.

The functional and non-functional requirements of the proposed system were outlined, along with anticipated challenges including ambient noise contamination in audio signals, dataset imbalance, and model generalization across diverse recording environments. Potential future extensions of the project were also identified and included in the presentation.

The prepared presentation was reviewed internally before being presented to the project guide and Head of Department during the Zeroth Review. Feedback and suggestions received during the review will be incorporated in the subsequent phase of the project.